

How much storage does a renewable system need?

Storage and flexibility

These solutions are written to be read after the workshop. Each question is worked in full, in the papers' own notation where the paper has one and in that of the lecture note *Energy System Models* (ESM below) where the machinery is ESM's. Reading questions are answered with the paper's own words and page numbers, so that you can check what you found against the text.

1 Why storage needs are two orders of magnitude below Sinn's

Part 1 · Reading Zerrahn et al.

This part reads sections 1–4 of the paper figure by figure. Its one analytical device is the residual load duration curve of §3.4, the paper's Fig. 4, repeated here as Figure 1; the other figures are the paper's, and the page numbers say where to find them.

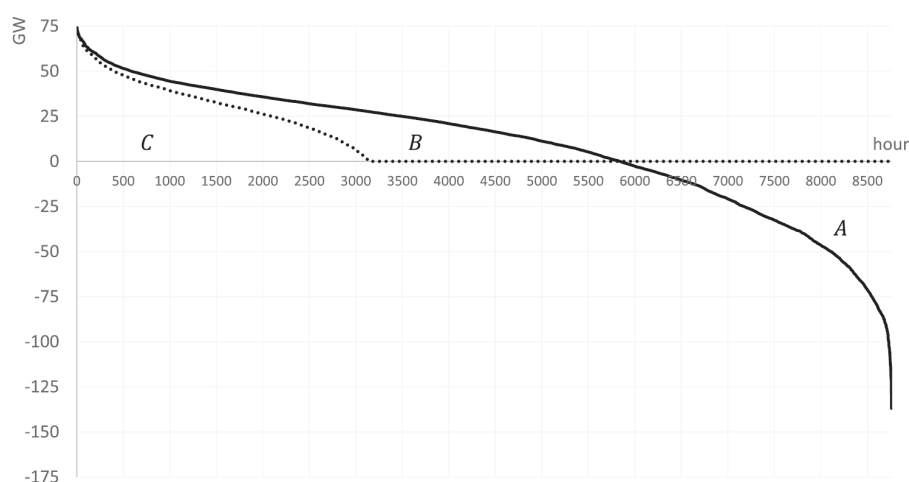


Figure 1: The paper's Fig. 4 (p. 265): Germany's residual load duration curve at 80% renewables, 2014 data. Solid: residual load; dotted: residual load after storage use, sorted again. Zerrahn et al. (2018), CC BY 4.0.

1. The claim

Sinn's claim. A system with a high share of wind and solar must store every surplus MWh, and the storage this takes grows so fast with the renewable share that it “could impede the further expansion of variable wind and solar power” (p. 260). The tanks he

finds are far beyond anything that exists (at 50% renewables, some fifty times Germany's pumped hydro).

The two assumptions (p. 260). First, Sinn “only considers two extreme cases in which either all surplus energy is stored or none”: so either storage needs are excessive or an excessive share of renewable energy is unused, and nothing in between is looked at. Second, he “does not explicitly consider an economic efficiency perspective”: his objective is to minimise the storage energy capacity subject to the renewable share, where an efficient solution would minimise the cost of reaching that share and trade storage off against renewables that are curtailed at times and against other assets. Section 4 of the paper relaxes the first assumption inside Sinn's own framework, section 5 the second.

Fig. 1. The figure sets every study in the literature against Sinn: storage in percent of annual demand, on a logarithmic scale, against the renewable share. Sinn's line lies far above all the others (by at least an order of magnitude), and the data labels say why: they are each study's curtailment, and every study below Sinn's line throws some renewable energy away, while Sinn throws none. So the figure already contains the diagnosis. It also shows that the field agreed on moderate storage needs before this paper; the paper's contribution is to show *which* assumption produces the outlier, not to find a new number.

2. Sinn's rule

The simulation (§3.2, p. 263). A spreadsheet on open data: a year of hourly German demand and wind and solar capacity factors, from Sinn's base year 2014. The renewable series is scaled up until renewables meet the share α of annual demand; renewables serve demand in the hour they are produced or, through the storage, in a later hour, and “some unspecified conventional technology” supplies the rest. The tank is the smallest for which the level never goes negative, Sinn's “empty in at least one hour”, with the level equal in the first and the last hour of the year “to avoid free lunch”. One run per share gives Fig. 2.

How the storage is used. By Sinn's heuristic: “if renewable generation exceeds demand in an hour, electrical storage takes up the surplus. It is released as soon as demand net of renewable generation is positive again” (p. 263). Every surplus MWh is stored and none is curtailed. The storage has only an energy capacity, with no limit on how fast it charges; that limit, the MW, is what the paper adds in §4.1. And the rule is myopic: it looks neither at prices nor ahead, but charges when there is surplus and discharges at the first hour of deficit.

Are the results realistic? As a replication, yes: Fig. 2 reproduces Sinn's numbers almost exactly. As a description of what a renewable system needs, no, for three reasons the paper gives.

- *Scale.* The tanks are orders of magnitude beyond anything built: Germany’s pumped hydro would integrate only about 30% renewables under the rule (p. 263). And countries with high shares of wind run without such storage (p. 262; Ireland, footnote 10).
- *Fragility (§3.3, Fig. 3).* The same rule on another year’s weather gives a very different tank (at 50%, less than a fifth of 2014’s with 2015 data). A number that moves this much with the weather year is a property of one year’s weather sequence, not of the renewable share.
- *The corner.* Every other study in Fig. 1 finds far less, and all of them curtail. The rule answers the question “how large must a tank be to waste nothing”, which no planner asks.

So the numbers are right for the question and the question is wrong, which is what sections 4 and 5 go on to show.

3. The residual load duration curve

The curve and its areas (§3.4, pp. 264–265). Residual load is demand minus renewable feed-in in each hour: what conventional plants or storage must serve. The residual load duration curve sorts the hours of the year by it, from highest to lowest. On Figure 1, area A, below the axis on the right, is the surplus: the hours in which renewables exceed demand, and the energy the storage takes up. Area B, above the axis, is where the stored energy goes back to meet positive residual load; it “equals the size of area A reduced by the storage’s efficiency losses”,

$$B = \eta_c \eta_d A \quad (1)$$

(about three quarters of A at the paper’s efficiencies). Area C is the residual load left for other generators, $1 - \alpha$ of annual demand by construction. The dotted line is residual load after storage use, sorted again on its own.

How the curve treats time. It does not: the hours are sorted by residual load, and the paper says it “dissolves the temporal sequence of hours during the year” (p. 265). That matters for the tank, because the tank must hold whatever has been put in and not yet taken out, which is a question about the *order* of the hours. Take the hint’s case: a surplus that arrives in one ten-hour run needs ten times the tank of the same surplus arriving in ten single hours, each followed by an hour of deficit that empties the tank, and area A is the same in both. So the curve shows how much energy passes *through* the tank in a year, not how large the tank must be; a tank that cycles often passes many times its own size. Under Sinn’s rule the tank is set by the longest contiguous surplus run, and §4.2 (p. 267) says exactly this: “extended periods of renewable surpluses in contiguous hours

determine the maximum energy capacity of the storage, and not single periods with the highest surplus generation”. Fig. 3 is the same point in data: the years that need small tanks had shorter runs, not smaller surpluses.

4. Storage with power-oriented curtailment

The idea, on Fig. 5 (§4.1, pp. 265–266). The point of Fig. 5 is that allowing a small amount of curtailment lowers storage requirements a great deal. The rule throws away every surplus above a threshold in MW, stores the rest, and keeps Sinn’s myopic operation otherwise. What it throws away, area D, is the tip of the curve: a few hours with extreme surpluses, which carry little energy but would otherwise set how fast the storage must be able to charge (in Fig. 5, 5% of the energy takes some 70% off the peak). The threshold “can be interpreted as the power capacity of a storage”, the energy it can take up per hour, as against the energy capacity it holds in total, and “this distinction is an essential characteristic of any electric storage technology and missing in Sinn”. For pumped hydro the MW are the pumps and turbines, the MWh the basin.

Figs. 6 and 7: storage and curtailment. Curtailment and storage are substitutes: in Fig. 6 the tank falls at every renewable share as the curtailment budget grows. The two intercepts of each line are Sinn’s two corners, all surplus stored on the vertical axis and no storage at all on the horizontal one, so each line is the space between them that Sinn never looked at, and a small step away from the vertical axis already cuts the tank sharply (at 50%, 5% curtailment takes it from 2.1 to 0.3 TWh). The decline is “close to linear though somewhat convex” (p. 265): each further percent of curtailment buys a similar reduction. Fig. 7 is the price of the substitution. To meet the same share with some energy thrown away, more renewable capacity is needed, but only a little more (a few percent). And curtailment does not raise backup needs: the left end of the curve is untouched in Figs. 4 and 5, and footnote 19 explains that it could only be otherwise if long surplus runs were followed directly by the highest residual loads, which the data do not show.

5. Storage with energy-oriented curtailment

The idea, on Fig. 8 (§4.2, pp. 267–268). Keep the myopic rule, but curtail “if and only if the storage is fully loaded”: charge while there is room in the tank, spill otherwise. The rule aims at the energy capacity, the MWh of basin, rather than the MW. Fig. 8 curtails the same amount of energy as Fig. 5 (the text on p. 267 says “area D in Fig. 6”, a slip for Fig. 5), but in other hours: not at the tip of the curve but in a sliver along the whole surplus side, in the hours “in which surpluses trigger the highest storage requirements; these are not necessarily the hours with the highest surplus generation”

(p. 267). Those are the hours at the end of the long runs, when the tank would otherwise have to keep growing. The rule still needs no foresight.

Fig. 9 against Fig. 6. Same axes, same shares, same curtailment budgets, and a very different shape: the decline is “much more convex” (p. 268). The first percent or two of curtailment now do most of the work, and a small budget takes the tank two orders of magnitude below Sinn’s, to less than the pumped hydro Germany already has (at 50% and 5% curtailment, 19 GWh against 0.3 TWh in Fig. 6). Question 3 says why. The tank is set by the long contiguous surplus runs, not by the tallest hours. The energy-oriented rule spends its whole budget on the hours that set the tank, at the end of those runs; the power threshold spends most of it on tall spikes outside them, so it cuts the MW fast and the MWh slowly.

What the two figures say together. Both rules keep Sinn’s objective, the smallest tank, and change only his corner, and that alone takes two orders of magnitude off. The paper’s reason why a planner would stop there too is on p. 261: “in equilibrium, the marginal effects of adding another unit of storage and adding another unit of renewable generation that gets curtailed at times are equal.” The last MWh of a tank big enough for every surplus is filled only at the end of the longest run, a few hours a year, while a few more turbines whose output is spilled now and then deliver in every other hour. So near zero curtailment the marginal storage is far more costly per MWh delivered, and the planner curtails until the two are equal. Curtailment is then not waste but the cheaper way to buy the marginal MWh. Part 2 writes this in duals.

6. Perspective: perfect foresight versus heuristics

Where each rule delivers. ESM’s storage holds a MWh as long as its shadow value θ_h exceeds what the current price would pay, and θ_h is constant between hours where a bound binds, so the energy is released into the hours of highest price: on Figure 1 the extreme left, where residual load and the price peak. Sinn’s rule releases into the first hour with positive residual load after a surplus, and the paper (p. 264) says what that means: “nothing forces the storage to shift energy to hours with highest residual loads, that is, greatest scarcity, at the very left-hand side of the RLDC”, and “it is unlikely that an hour with very high residual load follows closely to an hour with renewable surplus generation”. So Sinn’s storage lands somewhere in the middle of the residual load duration curve. The figure draws B as a band along the whole positive branch only because footnote 16 re-sorts the hours after storage use; it does not say where the energy went. In the language of §5.2, the myopic rule reaches B_1 , the planner reaches B_2 and B_3 .

What foresight buys. The LP “requires the assumption of perfect foresight to opti-

mally schedule the release of energy from the storage” (p. 270), and what the scheduling earns is the two values named on p. 271: the arbitrage value, displacing the most costly fuel rather than the next fuel, and the capacity value, shaving the peak that would otherwise need a peaker. The myopic rule can deliver neither, which is why footnote 19 finds that it does not reduce backup capacity. Market reality, says footnote 29, “likely lies between the myopic and the perfect foresight cases”: “proficient market forecasts are essential for storage operators”. ESM’s §4.4 makes the same reservation about its own battery: part of the arbitrage value it computes is the value of knowing the week’s prices in advance, and §9 returns to it.

Stop point

The two assumptions: corner solutions only, and no cost objective. Sinn’s rule stores every surplus and releases it at the first positive residual load; A passes through the tank and $\eta_c \eta_d$ of it reaches B. The residual load duration curve loses the order of the hours, and the tank is set by the longest contiguous surplus run, not by area A. A few percent of curtailment cut Sinn’s tank by an order of magnitude under the power-oriented rule and by two under the energy-oriented one.

Stretch question

Sinn’s rule against the optimum. Without the battery: hour 1 spills 0.5 GW of wind at a price of 0; hour 2 is coal alone at 40; hour 3 has coal at capacity and gas interior at 80. Sinn’s rule charges $\bar{s}$ in hour 1 for nothing and sells $0.9 \bar{s}$ in hour 2 at 40, as soon as residual load is positive: 36 per MW of rating. The optimum holds the $0.9 \bar{s}$ through hour 2 and tops the tank up with $\bar{s}/9$ from coal at 40, so that it can deliver its full rating $\bar{s}$ in hour 3 at 80: revenue $80 \bar{s} - 40 \bar{s}/9 = 75.6$ per MW, twice the rule’s. The discharge rating is the binding constraint in hour 3, which is why it does not sell $1.8 \bar{s}$ there. The shadow value of stored energy: charging is interior in hour 2, so $\lambda_2 = \eta_c \theta_2$ and $\theta_2 = 40/0.9 = 44.4$; no tank bound binds between the hours, so $\theta_h = 44.4$ in all three. Check: the rating rents are $\bar{\phi}_1^c = \eta_c \theta - \lambda_1 = 40$ and $\bar{\phi}_3^d = \lambda_3 - \theta = 35.6$, and their sum is the 75.6 the MW earns, which is part 2’s investment condition. The rule reaches only B_1 , displacing coal’s fuel; the optimum reaches B_2 , gas’s fuel at 80, and would reach B_3 , gas’s capacity, if the peaker were a choice. The rule’s mistake is to value the tank at the next hour’s price rather than at the best hour ahead, which is question 6 in three hours.

2 The cost-minimising model

Part 2 · The planner

Everything in this part is in the paper’s §5 and in the problem, Lagrangian and first-

order conditions printed on the worksheet, which are repeated here.

The problem, for every hour h , with the storage level equal in the first and the last hour, in ESM's form: every bounded variable between its bounds, its multipliers on the right:

$$\begin{aligned}
\min \quad & \underbrace{i^{se} N^{se} + i^{sp} N^{sp}}_{\text{storage energy and power}} + \underbrace{\sum_c i^c N^c + i^r N^r}_{\text{plant and renewables}} \\
& + \underbrace{\sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out})}_{\text{variable costs}} \quad (1) \text{ total cost} \\
\text{s.t.} \quad & d_h = \sum_c G_h^c + G_h^r + S_h^{out} \quad (2) \text{ hourly balance} \quad (\lambda_h) \\
& 0 \leq G_h^c \leq N^c \quad (3a) \text{ plant capacity} \quad (\underline{\mu}_{c,h}, \bar{\mu}_{c,h}) \\
& \gamma_h N^r = G_h^r + S_h^{in} + C_h \quad (3b) \text{ renewable energy} \quad (\nu_h) \\
& G_h^r \geq 0, \quad C_h \geq 0 \quad \text{renewable use} \quad (\underline{\mu}_{r,h}, \underline{\omega}_h) \\
& S_h = S_{h-1} + \eta_c S_h^{in} - S_h^{out} / \eta_d \quad (4a) \text{ storage balance} \quad (\theta_h) \\
& 0 \leq S_h^{in} \leq N^{sp} \quad (4b) \text{ charging rating} \quad (\underline{\phi}_h^c, \bar{\phi}_h^c) \\
& 0 \leq S_h^{out} \leq N^{sp} \quad (4c) \text{ discharging rating} \quad (\underline{\phi}_h^d, \bar{\phi}_h^d) \\
& 0 \leq S_h \leq N^{se} \quad (4d) \text{ energy capacity} \quad (\underline{\psi}_h, \bar{\psi}_h) \\
& \sum_{c,h} G_h^c \leq (1 - \alpha) \sum_h d_h \quad (5) \text{ renewable share} \quad (\sigma) \\
& N^{se}, N^{sp}, N^c, N^r \geq 0 \quad \text{capacities} \quad (\underline{\beta}^{se}, \underline{\beta}^{sp}, \underline{\beta}^c, \underline{\beta}^r)
\end{aligned}$$

The Lagrangian, in ESM's convention: minus the cost, and each constraint as a non-

negative slack under its multiplier, lower bounds included:

$$\begin{aligned}
\mathcal{L} = & - \underbrace{\left(i^{se} N^{se} + i^{sp} N^{sp} + \sum_c i^c N^c + i^r N^r + \sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out}) \right)}_{\text{minus the total cost (1)}} \\
& + \underbrace{\sum_h \lambda_h \left(\sum_c G_h^c + G_h^r + S_h^{out} - d_h \right)}_{\text{(2) hourly balance}} + \underbrace{\sum_{c,h} \left(\bar{\mu}_{c,h} (N^c - G_h^c) + \underline{\mu}_{c,h} G_h^c \right)}_{\text{(3a) plant capacity}} \\
& + \underbrace{\sum_h \nu_h (\gamma_h N^r - G_h^r - S_h^{in} - C_h)}_{\text{(3b) renewable energy}} + \underbrace{\sum_h (\underline{\mu}_{r,h} G_h^r + \underline{\omega}_h C_h)}_{\text{renewable use}} \\
& + \underbrace{\sum_h \theta_h (S_{h-1} + \eta_c S_h^{in} - S_h^{out} / \eta_d - S_h)}_{\text{(4a) storage balance}} + \underbrace{\sum_h \left(\bar{\phi}_h^c (N^{sp} - S_h^{in}) + \underline{\phi}_h^c S_h^{in} \right)}_{\text{(4b) charging rating}} \\
& + \underbrace{\sum_h \left(\bar{\phi}_h^d (N^{sp} - S_h^{out}) + \underline{\phi}_h^d S_h^{out} \right)}_{\text{(4c) discharging rating}} + \underbrace{\sum_h \left(\bar{\psi}_h (N^{se} - S_h) + \underline{\psi}_h S_h \right)}_{\text{(4d) energy capacity}} \\
& + \underbrace{\sigma \left((1 - \alpha) \sum_h d_h - \sum_{c,h} G_h^c \right)}_{\text{(5) renewable share}} + \underbrace{\underline{\beta}^{se} N^{se} + \underline{\beta}^{sp} N^{sp} + \sum_c \underline{\beta}^c N^c + \underline{\beta}^r N^r}_{\text{capacities}}
\end{aligned}$$

The first-order conditions, each set to zero, with complementary slackness on every bound:

Capacities. One condition for each capacity: minus its annuity, the multipliers of the constraints that capacity appears in, summed over the year, and the multiplier on its own lower bound.

$$\begin{aligned}
N^{sp} : & -i^{sp} + \sum_h \left(\underbrace{\bar{\phi}_h^c}_{(4b)} + \underbrace{\bar{\phi}_h^d}_{(4c)} \right) + \underbrace{\underline{\beta}^{sp}}_{N^{sp} \geq 0} = 0 & N^{se} : & -i^{se} + \sum_h \underbrace{\bar{\psi}_h}_{(4d)} + \underbrace{\underline{\beta}^{se}}_{N^{se} \geq 0} = 0 \\
N^c : & -i^c + \sum_h \underbrace{\bar{\mu}_{c,h}}_{(3a)} + \underbrace{\underline{\beta}^c}_{N^c \geq 0} = 0 & N^r : & -i^r + \sum_h \gamma_h \underbrace{\nu_h}_{(3b)} + \underbrace{\underline{\beta}^r}_{N^r \geq 0} = 0
\end{aligned}$$

Generation. One condition for each use of an hour's energy: the price λ_h against what conventional and renewable output cost at the margin, and the value ν_h of renewable energy against curtailment.

$$\begin{aligned}
G_h^c : & -v^c + \lambda_h - \underbrace{\sigma}_{(5)} - \underbrace{\bar{\mu}_{c,h} + \underline{\mu}_{c,h}}_{(3a)} = 0 & G_h^r : & \lambda_h - \nu_h + \underbrace{\underline{\mu}_{r,h}}_{G_h^r \geq 0} = 0 \\
C_h : & -\nu_h + \underbrace{\underline{\omega}_h}_{C_h \geq 0} = 0
\end{aligned}$$

Storage operation. One condition for each storage variable, with θ_h , the multiplier on the storage balance (4a), the value of a MWh in the tank at the end of hour h . In the last hour the cyclic condition takes the place of θ_{h+1} .

$$\begin{aligned}
S_h^{in} : \quad & -v^s - \nu_h + \eta_c \theta_h - \underbrace{\bar{\phi}_h^c + \underline{\phi}_h^c}_{(4b)} = 0 & S_h^{out} : \quad & -v^s + \lambda_h - \theta_h/\eta_d - \underbrace{\bar{\phi}_h^d + \underline{\phi}_h^d}_{(4c)} = 0 \\
S_h : \quad & -\theta_h + \theta_{h+1} - \underbrace{\bar{\psi}_h + \underline{\psi}_h}_{(4d)} = 0
\end{aligned}$$

1. The planner's problem

The problem in words. The planner minimises the annual cost of electricity, (1): the annuities of four capacities, storage energy N^{se} , storage power N^{sp} , conventional plant N^c for base and peak, and renewables N^r , plus the variable cost of every conventional MWh and of every MWh charged or discharged. It chooses those capacities and the hourly dispatch; in the paper's convention (p. 269) every upper-case Roman letter is a choice and everything in lower case or Greek is data. Constraint (2) is the hourly balance: demand is met by conventional generation, renewable generation and discharge. (3a) keeps each conventional plant within its capacity. (3b) says where the weather's renewable energy goes: to load, into the storage, or to curtailment, which is free. (4a)–(4d) are the storage: the level moves with charge and discharge net of losses, neither flow exceeds the rating, the level never exceeds the tank, and the year ends where it began. (5) is the policy: conventional energy may be at most the share $1 - \alpha$ of annual demand. The result is a long-run equilibrium under perfect competition, the paper says (p. 269), and it needs perfect foresight (footnote 29).

The multipliers. Each multiplier is a price. On the equalities, which are free: λ_h on the balance (2) is the price of a MWh in hour h ; ν_h on (3b) is the value of a MWh of renewable energy in hour h ; θ_h on (4a) is the value of a MWh in the tank at the end of hour h . On the upper bounds: $\bar{\mu}_{c,h}$ on (3a) is the capacity rent of plant c in hour h , positive only when the plant runs flat out; $\bar{\phi}_h^c$ and $\bar{\phi}_h^d$ on (4b)–(4c) are the rating rents, positive only when the storage charges or discharges flat out; $\bar{\psi}_h$ on (4d) is the full-tank rent, positive only when the tank is full; and σ on (5) is the shadow price of the target, positive only when the target binds. On the lower bounds, the underlined multipliers are positive only when their variable sits at zero, and then they say by how much it would lose to raise it: $\underline{\psi}_h$ when the tank is empty, $\underline{\phi}_h^c$ when the storage does not charge, $\underline{\omega}_h$ when nothing is curtailed, $\underline{\beta}$ when a capacity is not built. All of them must be nonnegative because, as in ESM, each inequality enters the Lagrangian as a slack that is nonnegative when it holds, and the planner maximises minus the cost: relaxing an inequality can only help, so its

price cannot be negative. Complementary slackness then says which multipliers are zero in a given hour: every bound that does not bind.

The capacity conditions, and σ . Each capacity is built until the hours in which its constraint binds pay its annuity. For a capacity that is built, its lower bound does not bind, $\underline{\beta} = 0$, and its condition says that the annuity equals the rents (for storage power, $i^{sp} = \sum_h (\bar{\phi}_h^c + \bar{\phi}_h^d)$). A capacity that is not built has $\underline{\beta} > 0$: its rents fall short of its annuity.

- N^{sp} : a MW of pumps is paid for by the hours in which the storage charges or discharges flat out, and by nothing else. It collects both rents because the paper assumes identical charging and discharging capacity (footnote 22): one inverter caps both flows.
- N^{se} : a MWh of basin is paid for by the hours in which the tank is full.
- N^c : a MW of plant is paid for by the scarcity hours in which it runs flat out. This is ESM's zero-profit condition for a plant in the long run.
- N^r : a MW of turbines is paid for by the value of its output over the year, $\sum_h \gamma_h \nu_h$. The hint's three conditions say what ν_h is. In an hour when renewables serve load, $G_h^r > 0$, so $\underline{\mu}_{r,h} = 0$ and $\nu_h = \lambda_h$, the price. In an hour when they fill the tank, $S_h^{in} > 0$, so $\underline{\phi}_h^c = 0$ and $\nu_h = \eta_c \theta_h - v^s - \bar{\phi}_h^c$, the value of stored energy net of the loss and the operating charge. In an hour when some are curtailed, $C_h > 0$, so $\underline{\omega}_h = 0$ and $\nu_h = 0$; in every other hour $\nu_h = \underline{\omega}_h \geq 0$. So a MW of turbines is paid for by the hours its output is not curtailed. The planner builds turbines whose output is sometimes worth nothing because the other hours pay for them: a MW that pays its way in most hours of the year is not made unprofitable by being idle in a few.
- G_h^c : when a conventional plant runs, but below capacity, both its bound multipliers are zero and $\lambda_h = v^c + \sigma$. The target acts as a tax of σ per MWh of conventional energy: the plant enters the merit order at its variable cost plus σ . It is a carbon price in disguise, denominated per MWh instead of per tonne, and zero whenever the target does not bind.

Read together, the four capacity conditions are the sentence of §6 (p. 273): in the optimum the marginal cost of expanding storage, renewables that are curtailed at times, and conventional capacity is equal, each per unit of what it delivers towards serving load under the target.

The storage's two capacity conditions say that its rents pay for it, but not yet what those rents are. Combining the operating conditions answers that in prices, and the key object is θ_h .

What θ_h measures. θ_h , the multiplier on the storage balance (4a), is the value to the system of one more MWh in the tank at the end of hour h , in EUR/MWh: the shadow price of stored energy, which ESM calls the water value. Every other multiplier prices a flow in one hour; this one prices a stock carried from one hour to the next. The condition for S_h says how it moves through the year:

$$\theta_{h+1} = \theta_h + \bar{\psi}_h - \underline{\psi}_h. \quad (2)$$

While the tank is neither full nor empty, both bound multipliers are zero and $\theta_{h+1} = \theta_h$: a MWh can then be carried freely into the next hour, so a MWh now and a MWh an hour later must be worth the same. The value jumps only where carrying is blocked: up across an hour with a full tank, by $\bar{\psi}_h$, and down across an hour with an empty one, by $\underline{\psi}_h$. Prices pin its level. In an hour when the storage charges but below its rating, both charging multipliers are zero and the condition for S_h^{in} gives $\nu_h = \eta_c \theta_h - v^s$; in an hour when it discharges below its rating, the condition for S_h^{out} gives $\lambda_h = \theta_h / \eta_d + v^s$. In the surplus hours in which the storage charges, renewables also serve load, $G_h^r > 0$, so $\underline{\mu}_{r,h} = 0$ and the condition for G_h^r gives $\nu_h = \lambda_h$ (both are zero when some of the surplus is curtailed). So θ_h is the price of the hours the storage trades in, net of its losses, and ESM's arbitrage band follows: charge when $\lambda_h < \eta_c \theta_h - v^s$, discharge when $\lambda_h > \theta_h / \eta_d + v^s$, hold in between.

The rating rents are price gaps. With $\nu_h = \lambda_h$, the conditions for S_h^{in} and S_h^{out} read

$$\bar{\phi}_h^c - \underline{\phi}_h^c = \eta_c \theta_h - \lambda_h - v^s, \quad \bar{\phi}_h^d - \underline{\phi}_h^d = \lambda_h - \theta_h / \eta_d - v^s. \quad (3)$$

The right-hand sides are the distance between the price and the edges of the arbitrage band, and the multipliers sort the hours by where the price falls. When the price is below the charging edge, stored energy is worth more than the price, the storage charges flat out, and $\bar{\phi}_h^c$ is the gap: the rent the rating collects because it stops the storage from buying more. The gap is largest in curtailed hours, where $\lambda_h = 0$ and the whole value of stored energy is rent. When the price is above the discharging edge, the storage discharges flat out and $\bar{\phi}_h^d$ is the gap between the price and the value of the energy taken out of the tank. Inside the band both rents are zero, and the lower-bound multipliers are positive in the hours the storage does not charge or does not discharge. So the condition for N^{sp} says that a MW of pumps is paid by the gaps between the price and the value of stored energy, in the hours when the rating binds: one more MW would buy one more MWh in each hour the charging rating binds and sell one more in each hour the discharging rating binds.

The full-tank rent is a jump in θ_h . In an hour with a full tank the empty-tank multiplier is zero, so by (2) $\bar{\psi}_h = \theta_{h+1} - \theta_h$: the full-tank rent is the jump in the value of

stored energy across the hour. Follow one fill of the tank, from the last hour in which the storage charges freely, at price λ^{in} , to the first in which it discharges freely, at price λ^{out} . Before the full hours $\theta = (\lambda^{in} + v^s)/\eta_c$, after them $\theta = \eta_d(\lambda^{out} - v^s)$, and in between it moves only across the full hours, so

$$\sum_{\text{full hours}} \bar{\psi}_h = \eta_d(\lambda^{out} - v^s) - (\lambda^{in} + v^s)/\eta_c. \quad (4)$$

This is the price spread one more MWh of tank would capture: it would take one more MWh at the fill price and release it at the later price, net of the losses. In a surplus run the tank fills from energy that would otherwise be curtailed, $\lambda^{in} = 0$, so the spread is roughly the later price net of the losses. The condition for N^{se} then says that a MWh of basin is paid by one such spread for each time the tank fills to the top, summed over the year; a tank that is never full earns no spread, and its last MWh could go at no loss.

Why the tank stops far below Sinn's. Put $i^{se} = \sum_h \bar{\psi}_h$ next to what part 1 found about the surplus. The full-tank rent is positive only in hours when the tank is full, and the last MWh of basin is full only at the end of the longest surplus runs. A skewed surplus arrives in a few such runs a year, so the last MWh of a tank big enough to hold every run collects a handful of rents: far too little to cover its annuity. The planner therefore stops building basin long before it holds every surplus, and the surplus the tank does not hold is curtailed. That costs nothing at the margin, because in those hours $\nu_h = 0$: the energy thrown away was worth nothing, and the turbines that produced it are paid for by their other hours. Sinn's tank is the one a planner would build if i^{se} were zero, since only then is a MWh of basin worth building for one full hour a year. That is the whole paper in two conditions: a tank is built until the hours in which it is full pay for it, and a skewed surplus means few such hours.

Perspective: against ESM. The remaining conditions are ESM's storage model of §4.1. S_h^{in} : the storage charges when the value of a MWh in the tank, net of the charging loss, covers the value of the renewable energy put in plus the operating charge; if it charges at its rating, the rating rent takes the difference. S_h^{out} : it discharges when the price covers the stored value grossed up for the discharging loss plus the operating charge, with the rating rent again taking the rest. These two are ESM's arbitrage band, with the renewable value ν_h in place of the price on the charging side because the paper's storage may only take renewable energy. S_h : the value of stored energy is constant from one hour to the next except where the tank is full, where it steps up by the full-tank rent $\bar{\psi}_h$, or empty, where it steps down by the empty-tank multiplier $\underline{\psi}_h$. The three conditions are ESM's storage conditions of §4.1 term for term, bound multipliers included, with ν_h for λ_h in the charging one. This is ESM's rule that θ_h moves only in hours where a storage constraint binds. The paper's (1)–(5) and ESM's storage model are therefore the same LP with three

additions; the table maps one onto the other:

Paper	ESM	Paper	ESM
S_h^{in}, S_h^{out}	g_h^c, g_h^d	G_h^c, N^c	$q_{g,h}, \bar{q}_g$
S_h	soc_h	$\gamma_h N^r$	$\gamma_{g,h} \bar{q}_g$
N^{sp}	$\bar{s}$	C_h	the slack in $q_{g,h} \leq \gamma_{g,h} \bar{q}_g$
N^{se}	$\bar{\delta} \bar{s}$	v^c, v^s	c_g, c^s
η^{in}, η^{out}	η_c, η_d	d_h	D_h

The three things the paper’s planner has that ESM’s model does not: every capacity is a *choice*, with an annuity i in the objective, where ESM holds the fleet and the battery fixed; the share constraint (5), where ESM prices carbon instead; and the storage may charge only from renewables (p. 270), which ESM’s single bus does not distinguish. Going the other way, ESM has what the paper’s §6 lists as left out: imports, a carbon price, and in §4.3 ramping. What carries over unchanged is everything about operation: the arbitrage band, the rule that θ_h moves only where a storage bound binds, and the duals table of §4.1, because none of the three additions enters the charging and discharging conditions. What is new is the investment layer on top of it, and the capacity conditions are that layer.

2. The residual load duration curve at the optimum

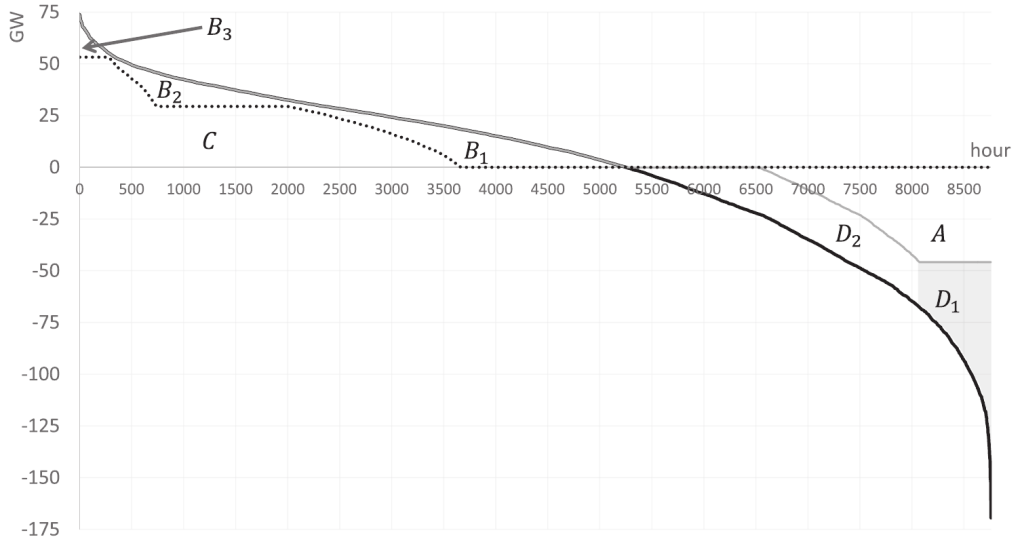


Figure 2: The paper’s Fig. 10 (p. 271): Germany’s cost-optimal solution at 80% renewables. Black: residual load; grey: after curtailment; dotted: after curtailment and storage, each sorted on its own. Zerrahn et al. (2018), CC BY 4.0.

The paper’s words (§5.2, pp. 270–271). The optimum uses both of part 1’s curtailment strategies at once, and sends the stored energy to three uses. D_1 and D_2 are the

curtailed renewable energy: D_1 is the surplus beyond a threshold “limited by the optimal storage power capacity”, the power-oriented kink; D_2 is “driven by energy-oriented renewable curtailment, targeted at limiting the storage energy capacity”. A is the stored surplus. Its release “economically balances three uses”: B_1 reduces generation and variable cost of the base technology; B_2 reduces variable cost of the peak technology, in the hours when residual load exceeds base capacity; and B_3 “replaces capacity entry, and respective cost, of the peak technology”, at the very top of the curve.

Reading the figure. Each flat part of Figure 2 is a capacity that binds. In hours of positive residual load the dotted line is what conventional plants still generate, so its flat parts are plant capacities: levels conventional output cannot pass. The storage’s own limit shows on the grey line, on the surplus side. From left to right:

line	level	what binds	the storage	area
dotted, top left	53.5 GW	base and peak flat out	covers all above	B_3
dotted, middle	29.4 GW	base flat out, peak off	discharges down to base	B_2
dotted, right	0	no conventional output left	covers all residual load	B_1 before it
grey, far right	−46 GW	the charging rating	charges flat out	D_1 below
grey, surplus hours	up to 0	a full tank	stores nothing more	D_2

The top-left plateau is the capacity value: without storage the peak hours would need more peaker capacity, and the paper says the dotted line there marks the demand that “additional generation capacity would have to satisfy” (p. 271). The middle plateau is the arbitrage value against the peaker: in those hours the storage discharges exactly down to base capacity, so the peaker stays off. Between the middle plateau and zero the dotted line slopes down, and there the storage displaces base generation. On the surplus side, the grey line is flat at the charging rating, and everything below it is curtailed; to its left, in hours where the tank is full, the grey line lies above the black one, up to zero where the whole surplus is spilled.

Why the dotted line is flat rather than sloped: in an hour when the storage discharges below its rating, the condition for S_h^{out} sets the price at its discharge edge, $\lambda_h = \theta_h/\eta_d + v^s$. At a capacity kink, where one plant runs flat out and the next is off, the price can sit anywhere between the two plants’ costs (each plus σ), with the capacity rent $\bar{\mu}_{c,h}$ taking the difference. So every hour whose discharge edge falls in that range stops at the same level. The curve is sorted, so these hours come from different fills of the tank, each with its own θ_h .

Where the surplus goes. The surplus is the area between the black line and zero on the right. D_1 and D_2 are both curtailed; A , between the grey line and zero, is what the storage charges, since after storage nothing is left of the surplus (the dotted line is at zero there). So the surplus is $D_1 + D_2 + A$, curtailment is $D_1 + D_2$, and what comes back to

meet residual load is

$$B_1 + B_2 + B_3 = \eta_c \eta_d A. \quad (5)$$

The two values. An *arbitrage value*, when stored surplus replaces other generators' variable costs, which is B_1 and B_2 ; and a *capacity value*, when it replaces plant that would otherwise have to be built for the residual load peaks, which is B_3 (p. 271). The paper adds that its storage understates both, because it may only take renewable energy, and that in reality nothing prohibits charging from conventional plants; part 3 tests that remark.

Beyond the question: why both strategies. “If capacity entry is costly with respect to both storage power and storage energy, the optimal solution combines both channels analyzed in Section 4” (p. 270). Question 1 says why in duals. The rating is built until the rating rents pay for it, and whatever surplus exceeds it in any hour is curtailed: D_1 , the power-oriented rule with the threshold set by cost rather than by a curtailment budget. The tank is built until the full-tank rents pay for it, and when a long surplus run fills it the rest of the run is curtailed: D_2 , the energy-oriented rule. Each capacity has its own price and its own rent, so each is cut off at its own margin.

3. The numbers

Germany (§5.3, pp. 271–272). Optimal storage is moderate and short at every share. At 50% it is a daily store, “a typical short-to-medium-term storage to compensate diurnal fluctuations” (35 GWh and about 6 GW, some six hours). At 90% it is much larger but still holds less than a day at full power (about 1,080 GWh and 58 GW, some 19 hours). Curtailment rises with the share, from about 5 to about 22% of the renewable energy. The conclusion: “no need for a true long-term storage arises, that is, storing energy for weeks or months.”

Why E/P rises, and why curtailment rises. Storage energy “grows much faster than the optimal storage power capacity” (p. 271) because the two are paid for by different hours, which is the hint. The rating is paid by the hours at the rating, and the height of the surplus spikes grows in proportion to the renewable fleet; the tank is paid by the hours it is full, and what grows with the share is the length of the surplus runs and the depth of the calm spells the tank must bridge. So energy grows faster than power, and the duration lengthens. This is why the paper insists on pricing “rather inexpensive storage energy and rather expensive storage power separately”. Curtailment rises because “the economics of renewable electricity provide no reason why curtailment should be avoided” (p. 271): as the target tightens, the extra surplus arrives in ever longer runs that a tank

would have to be ever larger to hold, so overbuilding and spilling becomes the cheaper way to deliver another percent.

Beyond the question: no storage at low shares. Fig. 11 shows no storage at all at low shares (below about 35%). There the planner meets the target with turbines alone: they are cheaper per MWh than conventional fuel, so they are built until their last MW is curtailed often enough for its captured value $\sum_h \gamma_h \nu_h$ to fall to its annuity, and until then a tank has nothing to hold. Part 3 finds the same effect for DK1, at a much higher share because ESM’s costs are later and lower.

4. Power-to-x

What is added, and what happens (§5.4). A large new load whose timing is free, “for instance a fleet of electric vehicles” or electrolyzers, entered in the renewable balance (3b’) with an hourly capacity (6a) and an annual energy (6b), and served entirely by additional renewables. At moderate shares it largely replaces storage and curtailment, because it consumes the surplus in the hours it occurs (Fig. 12: at 50%, storage falls from 35 to 4 GWh and curtailment to under 1%). At high shares it leaves storage unchanged, because the new load “can be completely satisfied from renewable surplus generation that would otherwise be curtailed”: it eats surplus the planner was already throwing away, and the tank was never built for that energy.

The sentence. A load that can be moved to the surplus hours consumes area A where it occurs, which is what a storage does with a tank in between: a shiftable load is a storage without the tank, and without the tank’s losses and cost, but it can only take energy, never give it back.

Beyond the question: Appendix D. The substitution can reverse. Appendix D (Fig. D.1) varies how many hours a year the new load must run: when it must run most of the year, it stops looking like a surplus-absorber and starts looking like more demand with a profile that does not match renewable availability, which forces disproportionate renewable expansion and with it more curtailment and more storage.

Stop point

A MW of pumps is paid for by the hours at the rating, $i^{sp} = \sum_h (\bar{\phi}_h^c + \bar{\phi}_h^d)$; a MWh of basin by the hours the tank is full, $i^{se} = \sum_h \bar{\psi}_h$; a MW of turbines by the hours its output is not curtailed, $i^r = \sum_h \gamma_h \nu_h$; and a conventional plant enters the merit order at $v^c + \sigma$. A skewed surplus fills the last MWh of basin too rarely to pay for it. Germany's optimal storage is short and moderate at every share (35 GWh, six hours, at 50%), never seasonal, and a flexible new load largely replaces it at moderate shares.

Stretch question

Appendix C. Lithium-ion is cheap in MW and costly in MWh, with a higher round-trip efficiency. In the conditions: the sum of full-tank rents must now reach a higher annuity, so the planner builds fewer MWh of tank and lets long runs spill; the sum of rating rents must reach a lower one, so the rating is relatively cheap and the E/P ratio falls; and with costly MWh replaced by spill, curtailment rises. The appendix confirms all three (at 50%, about a third of the MWh of the pumped-hydro case), and total cost is slightly higher because the cheaper option is gone. Qualitatively nothing changes, which is the paper's third finding there. The §6 point is that storage is a family indexed by duration: batteries for a few hours, pumped hydro for longer, power-to-gas for longer still, each with its own pair of prices for MW and MWh, and a portfolio of them “tends to be smaller and cheaper” than any one.

3 What the paper's model says about Denmark

Part 3 · The planner on DK1

This part was not worked in class; it is an optional exercise, worked here in full. The evidence is the paper's LP (1)–(5) on DK1's 2024 load and weather, with ESM's 2050 costs and its lithium-ion battery, CCGT and OCGT as base and peak. Everything below is read off the worksheet's Table 1; its figures are repeated here, with the paper's areas marked, and one more is added.

1. DK1 against the paper’s conclusions

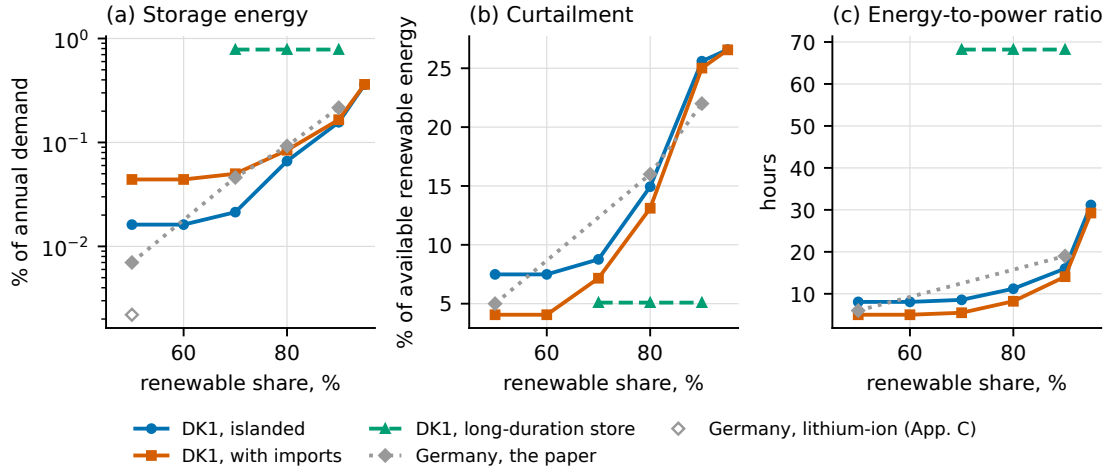


Figure 3: DK1 against the paper’s Germany: (a) storage energy as a share of annual demand, the axis of the paper’s Fig. 1; (b) curtailment; (c) the energy-to-power ratio. Germany’s points are the paper’s numbers.

DK1 at 80%. Relative to its size, DK1 needs about as little storage as Germany, and far less than Sinn. Its 80% tank is a few hours of average load (15.97 GWh, which is 0.066% of annual demand or 5.8 hours of average load, against Germany’s 0.092%). The share of annual demand is the comparison to make, because Germany’s demand is some twenty times DK1’s; that is why the paper’s Fig. 1 divides by it. On that axis DK1 sits a little below the paper’s curve and rises as steeply with the share, as the paper’s Fig. 1 says storage does everywhere in the literature. Against Sinn’s line it is one and a half to two orders of magnitude lower, the paper’s “up to two orders of magnitude” (abstract).

Curtailment. DK1 curtails about as much as Germany, and for the same reason: spilling part of the surplus is what keeps the tank small (14.9% at 80%, against Germany’s 16%, or 21% with lithium-ion costs in Appendix C, the closer comparison since DK1’s storage is lithium-ion too). Curtailment also rises with the share along the same path as Germany’s, and Figure 3(b) lays the two curves on top of each other. The paper’s conclusion holds on DK1 in both halves. DK1 curtails a little less than Germany-with-batteries, which fits its smoother, offshore-heavy profile.

The long-duration rows. A much cheaper tank buys days of storage, not weeks. With the cells at a tenth of their cost, storage replaces curtailment, which is the paper’s mechanism run backwards: the planner builds a far larger tank, curtails much less, and reaches a high share without being forced. But the tank still empties in under three days at full power (an E/P ratio of 68 hours). The paper’s “true long-term storage” is “storing

energy for weeks or months” (p. 271), and even a tank this cheap is not built for that. The reason is the N^{se} condition: a seasonal store fills once a year, so its last MWh must recover its whole annuity from one cycle. Cheaper cells buy longer runs, days instead of hours, but the weeks-long runs still fill the last MWh too rarely.

Beyond the question: why DK1 lies where it does. Two of the paper’s own points. First, the mix (§6, fourth point): a large part of DK1’s renewable capacity is offshore wind, which the paper singles out as having “much smoother generation patterns and higher full-load hours” than onshore wind, and a smoother profile needs less storage per unit of demand. Second, the weather year (Appendix B): the paper finds large differences between base years and notes that 2014 is the worst of its five; DK1’s 2024 is one draw. A third reason is not the paper’s but ours: the storage is lithium-ion, cheap in MW and costly in MWh, and the paper’s Appendix C shows that this alone cuts the optimal tank to about a third. That is why DK1 sits below the pumped-hydro curve and near the open diamond in Figure 3(a).

2. The residual load duration curve for DK1

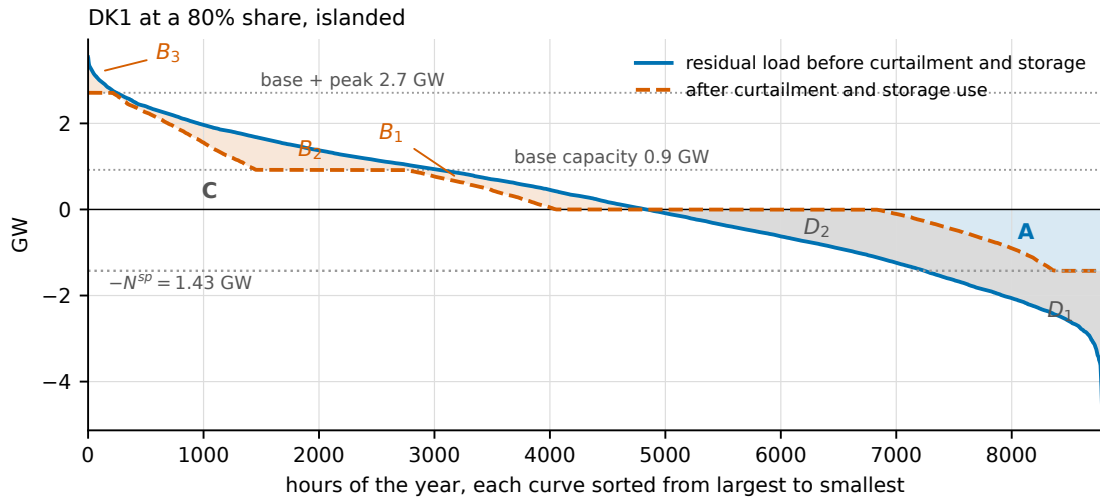


Figure 4: The worksheet’s residual load duration curve with the paper’s areas marked. Grey is curtailed (D_1 below the rating, D_2 above it), blue is stored (A), orange is displaced by discharge (B_1 base fuel, B_2 peak fuel, B_3 peak capacity), and C is what conventional plants still serve.

The three plateaus. The dashed curve in Figure 4 is conventional generation in deficit hours and minus the charge in surplus hours, sorted. Each flat part is a capacity that binds. From the right:

- the storage charging at its rating, the paper’s power-oriented kink (at -1.43 GW, the N^{sp} column of the table at 80%);
- zero, many surplus hours in which nothing is stored: the tank is full or the run too long, and the whole surplus is curtailed, the paper’s energy-oriented rule;
- on the left, the base plant running flat out with the peaker idle, the paper’s “horizontal part of the dotted line” at base capacity (p. 270), with a last plateau above it where both plants run flat out.

The five areas. DK1’s optimum has the same anatomy as the paper’s Fig. 10: both curtailment strategies, and all three uses of the stored energy. D_1 is the grey below the dotted line: surplus curtailed because the inverter is full. D_2 is the grey between the two curves above it, in the hours on the zero plateau: surplus curtailed because the tank is full. The blue between them is A, the stored energy. On the deficit side, discharge shaves residual load down to the base plateau, displacing the peaker’s fuel: B_2 . Where the dashed curve lies below the solid one and below the base plateau, discharge displaces the base plant’s fuel: B_1 . And at the very top, residual load before storage peaks above the conventional capacity the planner builds, so the storage has taken some peaker capacity out of the plan: B_3 , the capacity value. Most of the surplus is thrown away rather than stored (about two thirds), and the planner is content with that, because the marginal MWh of basin would fill in too few hours a year to pay for itself.

3. What the target costs

Unforced. $\sigma = 0$ at a 50% target means the share constraint does not bind: on ESM’s 2050 costs, renewables are cheaper per MWh delivered than the gas they replace, so the planner builds them well past 50% on cost alone (68%). The hint’s division shows it (a MW of the blend costs about 37 EUR per MWh it delivers, against 63 EUR/MWh of CCGT fuel alone). The planner builds turbines until their last MW is curtailed often enough for its captured value to fall to its annuity, which is part 2’s condition $i^r = \sum_h \gamma_h \nu_h$. The paper’s 2035 costs put the same crossing far lower; the difference is a decade and a half of cost decline.

Above the crossing. By part 2, σ is a tax on every conventional MWh, so the target is a carbon price in disguise, and dividing by the CCGT’s emissions per MWh turns it into one (at 80%, $\sigma = 41.9$ EUR/MWh; the CCGT emits $0.198/0.6 = 0.33$ t per MWh, so about 127 EUR/t). The price rises steeply towards the end: the last percent of renewables is the most costly, which is the marginal-cost reading of σ and the steepening of both panels of Figure 5.

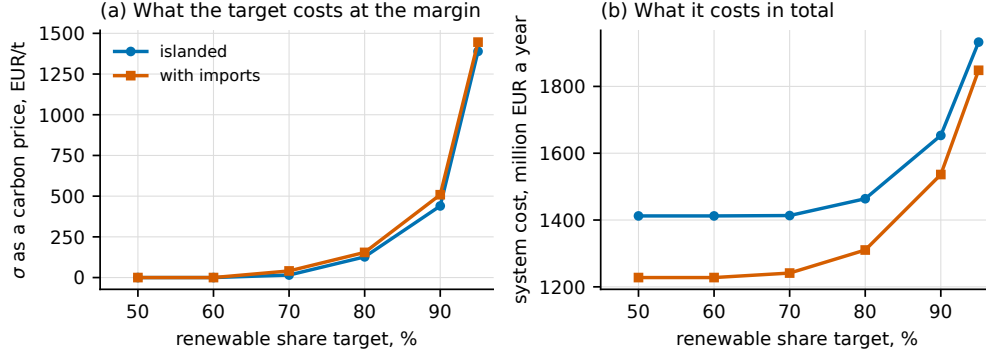


Figure 5: The share constraint’s multiplier σ , converted to a carbon price with the CCGT’s emission rate, and the system cost of the target. Both are flat where the target is slack and steepen at the end.

4. Perspective: §6 tested on DK1

Imports raise storage. A one-way import connection raises storage rather than lowering it, because it cannot take the surplus (20.33 GWh against 15.97 islanded at 80%). This is not the paper’s interconnection. Its §6 says geographical balancing smooths residual load, and its footnote 17 says part of area A “is likely to be exported”: the wires work on both sides of the residual load duration curve. ESM’s backstop works on one. DK1 may buy at the neighbours’ price and may not sell, so the surplus still has nowhere to go and the storage is still the only alternative to curtailment. What the wires do change is the deficit side: they replace the domestic fleet with an hourly price that carries the neighbours’ scarcity peaks, and a storage charged with free surplus and discharged into those peaks is worth more than one discharged against a flat OCGT. Two-way trade would do what the paper says; a one-way backstop does the opposite, and the modelling choice is the whole result.

Free charging changes nothing. The paper’s claim holds for its lignite-and-gas fleet, where a stored lignite MWh displaces far more costly gas. In DK1’s fleet both plants burn gas, so the spread between base and peak fuel is small, and with the share binding every conventional MWh also carries σ , so the round-trip loss is taxed too. Nor is the storage idle: its rating is committed to the surplus. So the restriction the paper imposes for convenience costs nothing here.

Beyond the question: duration, and ESM’s battery. DK1’s optimal storage is a daily-to-few-day store, like Germany’s (Figure 3(c), inside the paper’s range of six to nineteen hours). This is the place to set the paper against ESM. ESM’s §4.4 finds that even a very large fleet of 4-hour batteries barely moves the value of Danish wind, because the surpluses it would have to absorb last for days: against a week of surplus the battery

is full by the first afternoon. The paper’s planner meets the same weeks and does not store them either; it curtails them, and builds the tank for the daily cycle and the few-day runs that pay many times a year. The two are one statement seen from two sides: a fixed 4-hour battery cannot help wind because the runs are too long, and a planner free to size the tank will not build one long enough to try.

Stop point

DK1 builds 15.97 GWh at 80%: 5.8 hours of average load, 0.066% of annual demand, a little below the paper’s Germany at 0.092%. The planner reaches 68% unforced. At 80%, $\sigma = 41.9$ EUR/MWh is about 127 EUR/t.

Stretch question

Sinn on DK1, and the weather year. Sinn’s rule would put DK1’s storage an order of magnitude above the planner’s (3.23% of 24.1 TWh is about 780 GWh, twenty times the 37.77 GWh the planner builds at 90%), much as for Germany. The statistic is the length and energy of the longest contiguous surplus run, or more generally the autocorrelation of residual load: the paper attributes the base-year spread to “smoother patterns of residual load” (Appendix B), and part 1’s question 3 says why the residual load duration curve cannot see it. A year whose calm and windy spells last longest needs the largest tank, whatever its annual surplus. In DK1’s case that is a question about how long the North Sea weather systems sit, not about how much wind blows in the year.

References

Zerrahn, Alexander, Wolf-Peter Schill, and Claudia Kemfert (2018). “On the Economics of Electrical Storage for Variable Renewable Energy Sources”. In: *European Economic Review* 108, pp. 259–279.